An Executable Native Entrance for the TPC-15 Hard Packet

Canonical Tuple Enumeration, Symbolic Coefficient DAGs, and Scale-Uniform Completeness

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Abstract

TPC-119 and TPC-123 identify the complete native path archive as the first missing structural object in the fixed-shift route, but their programs test only finite coordinate models. We construct the first literal component of that object. For fixed $h_0 \neq 0$, rational $0 < \delta < 1/2$, a fixed smooth weight with a certified compact support envelope, and every integer X, a terminating generator enumerates exactly once every candidate tuple

$$(\ell, k, d), \qquad \ell > U, \quad k > V, \quad d \mid k, \quad d \leq V$$

in that envelope. The coefficient is retained as the symbolic expression

$$-\Lambda(\ell)\mu(d)r_Q(\ell k + h_0)W(\ell k/X);$$

no programmatic decision about the exact zero set of W, and no floating evaluation of this expression, is used. The generated sum is exactly the opened TPC-15 hard packet. Canonical tuple serialization, rather than a cryptographic hash, proves record uniqueness. The uniform enumeration and divisor opening are L0, and their attachment to the literal packet is L1. No downstream stage archive, positive L2 estimate, parity advance, or prime-pair theorem is proved.

1 The literal entrance and the decidability boundary

Fix $h_0 \in \mathbb{Z} \setminus \{0\}$, rational $\delta \in (0, 1/2)$, and

$$Q = \lfloor X^{1/2-\delta} \rfloor, \qquad U = V = \lfloor \sqrt{Q} \rfloor.$$

The TPC-15 packet is

$$B_{h_0,\delta}(X) = -\sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell)\beta_V(k)r_Q(\ell k + h_0)W(\ell k/X), \qquad \beta_V(k) = \sum_{\substack{d \mid k \\ d \leq V}} \mu(d) \tag{1}$$

[1, 2]. Let $\operatorname{supp} W \subset [a, b] \Subset (0, \infty)$, where $a, b \in \mathbb{Q}$ are part of a proof-carrying packet manifest.

For a general smooth function, deciding from finite source text whether $W(x)$ is exactly zero at an arbitrary rational point is not an appropriate archive primitive. We therefore enumerate an envelope rather than claim a minimal active set.

Definition 1.1 (Support-envelope native set)**.** For every integer $X \geq 2$, let

$$\widetilde{\mathcal{A}}_X = \left\{ (\ell, k, d) \in \mathbb{N}^3 : \ell > U,\ k > V,\ d \mid k,\ d \leq V,\ aX \leq \ell k \leq bX \right\}.$$

The coefficient attached to $\alpha = (\ell, k, d)$ is

$$c_X(\alpha) = -\Lambda(\ell)\mu(d)r_Q(\ell k + h_0)W(\ell k/X). \tag{2}$$

The set may contain zero coefficients, including those caused by an interior zero of W or by $\mu(d) = 0$. Retaining a zero column is harmless; deleting a column on the basis of an uncertified numerical zero would not be.

2 Uniform finite enumeration

Theorem 2.1 (Executable native entrance)**.** *There is one deterministic algorithm which, for every manifest in the scope above and every integer $X \geq 2$, terminates and outputs a finite list $\mathsf{Gen}_{133}(X)$ whose tuple projection is a bijection onto $\widetilde{\mathcal{A}}_X$. Its record count is*

$$|\widetilde{\mathcal{A}}_X| = \sum_{\substack{k>V \\ \exists\,\ell>U:\ aX\leq \ell k\leq bX}} \tau_{\leq V}(k)\#\{\ell > U : aX \leq \ell k \leq bX\},$$

where $\tau_{\leq V}(k) = \#\{d \mid k : d \leq V\}$.

Proof. The product restriction gives

$$\ell \leq \frac{bX}{V+1}, \qquad k \leq \frac{bX}{U+1}.$$

Thus both outer loops are finite. For each admissible pair (ℓ, k), enumerate the positive divisors of k not exceeding V, in increasing order. Every output satisfies the defining conditions. Conversely, each tuple in $\widetilde{\mathcal{A}}_X$ occurs when its unique ℓ, k, and d loop values are reached. No loop triple repeats, proving injectivity and the count. □

Corollary 2.2 (Exact opened packet)**.** *For every manifest in theorem 2.1,*

$$\boxed{B_{h_0,\delta}(X) = \sum_{\alpha\in\mathsf{Gen}_{133}(X)} c_X(\alpha).}$$

Proof. Open $\beta_V(k)$ in (1) and interchange finite sums. Terms with $\ell k/X \notin [a, b]$ vanish because of the certified support envelope. The remaining tuples are exactly $\widetilde{\mathcal{A}}_X$. □

3 Exact records without false numerical claims

The machine record stores (2) as an abstract syntax tree with nodes for Λ, r_Q, and W, together with an exact integer leaf equal to $-\mu(d)$. Thus the finite Möbius value is evaluated exactly, while no floating evaluation of the other factors is used. Conservation in later papers acts on the branch multiplier and does not require deciding the transcendental values $\log p$ or $W(\ell k/X)$.

Definition 3.1 (Canonical identity)**.** The mathematical identifier of a native record is the full serialized tuple

$$(X, h_0, \ell, k, d)$$

together with the packet-scope fields Q, U, V, W and the unique physical normalization. A SHA-256 digest is an integrity checksum, not a proof that two mathematical records are distinct.

The short field `native_id` used by the sample is only a scope-local convenience label. Canonicality is checked from the full $(\text{packet scope}, \text{native tuple})$ key before the digest is computed. In particular, changing a packet-scope field while keeping the same short label creates a different mathematical record.

Proposition 3.2 (Mutation stop certificates)**.** *For a declared finite output, each of the following is an exact certificate that the implementation does not instantiate theorem 2.1: a duplicated or missing tuple, an incorrect divisor constraint, a changed h_0, a second normalization, or a coefficient syntax tree different from (2).*

Proof. The first two violate the bijection. The remaining mutations violate the record definition or the packet scope. They stop that implementation, not every possible archive language. □

4 Certificate and claim boundary

The standard-library program `experiments/tpc133_native_entrance.py` computes Q, U, V by integer inequalities, enumerates a deterministic sample, checks its count through the transposed k-first census, and rejects deletion, duplication, wrong-shift, second-normalization, and coefficient-tree or support-envelope mutations. Default mode writes deterministic JSON and JSONL artifacts. The `-check` mode is read-only.

Statement	Status
Uniform support-envelope enumeration	Proved, L0.
Exact divisor opening of the TPC-15 packet	Proved, literal L1 entrance.
Minimal exact nonzero support of arbitrary W	Not claimed and not required.
Complete TPC-123 downstream stage archive	Not supplied.
Fixed-h_0 growing arithmetic saving	Not proved; no positive L2.

Stop versus not-testable. An explicit malformed record is STOP-DECLARED-ROUTE for that implementation. An absent support proof, weight definition, or packet manifest is NOT-TESTABLE. Neither outcome is a statement about the twin-prime conjecture.

Explicit exclusions. This paper proves no physical cover, determinant-energy bound, zero-mode saving, signed Möbius or Liouville estimate, endpoint below 1/400, Hardy–Littlewood asymptotic, prime-pair lower bound, or twin-prime theorem. It is the entrance of an archive, not its arithmetic exit.

References

[1] Liang Wang. Reconnecting the original fixed-shift hard packet. *Preprint*, 2026.

[2] Liang Wang. Fixed-shift local models within the square-root distribution range. *Preprint*, 2026.